

FINAL REPORT

Out-of-Wind-Tunnel Dynamic Testing of Propellers

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Abstract

The growth of the UAV industry in recent years has led to a resurgence in designing and testing propellers at low Reynolds numbers [1]. The advent of lightweight and higher quality brushless DC motors has made electric powered UAVs a cost-effective alternative to more expensive, louder, and less efficient jets of comparable power output. With applications ranging from the defense industry to commercial and recreational uses, there is an urgent need for UAV designers to fully characterize and predict the behavior of their electric propulsion systems prior to flight.

The goal of this project is to design and build a vehicle-mounted thrust stand for dynamic testing of propellers that can replace traditional wind tunnel testing methods. Throughout the course of this project, a car-mounted dynamic test stand was constructed that recorded the performance of the propeller and test environment. The data collected showed acceptable repeatability and qualitatively matched trends expected from literature data for propellers larger than 9 inches in diameter. By reducing the measured performance of the propellers into characteristic coefficients, the results were compared to equivalent wind tunnel tests performed by Deters and Selig [3]. It was observed that thrust measurements were consistently lower than wind tunnel baselines but that terms derived independent of the thrust measurements were equivalent at all airspeeds. As a result of these trends, it was concluded that the design of the dynamometer used for testing resulted in momentum losses behind the propeller that produced reductions in only the measured thrust values. To account for the momentum loss behind the propellers, the test stand can be redesigned to mitigate these effects and record data that is expected to match wind tunnel baselines.

Introduction

Although static conditions can be tested simply and effectively using any number of commercially available thrust stands, dynamic testing of propellers has historically been limited to large companies and research institutions with access to wind tunnels. Experimental results have shown that a wind tunnel test section must be at least five times larger than the diameter of the propeller to avoid boundary interference effects [2]. With this consideration in mind, it becomes difficult for even the largest universities to perform dynamic testing on propellers that exceed 10 inches in diameter.

As the ultimate goal of the project is to bring dynamic testing capabilities to a broader audience, the contents of this report are intended to enable the reader to recreate the results presented and perform their own testing. To this end, technical drawings, test procedures, and detailed cost estimates are provided as appendices.

Theory

The complex cross sections and twist distributions that define propeller geometries are reduced to two variables—the diameter, or tip-to-tip length; and pitch, the theoretical distance travelled in one revolution. For a propeller of diameter, D , and pitch, p , the design advance ratio, J_D , can be calculated according to Equation (1).

$$J_D = \frac{p}{D} \quad (1)$$

The design advance ratio defines the optimal operating conditions for a propeller. As a relationship between the rotational and freestream velocity experienced by the propeller, the design advance ratio expresses the relative velocity at which each cross section along the propeller has been designed to operate most efficiently [3, 1]. The true advance ratio of an operating propeller can be calculated according to Equation (2) where, V_∞ is the freestream velocity and n is the propeller's rotational speed.

$$J = \frac{V_\infty}{nD} \quad (2)$$

In order to compare different propellers, measured performance can be non-dimensionalized to obtain the thrust and power coefficients shown in Equations (3) and (4), respectively.

$$C_T = \frac{T}{\rho n^2 D^4} \quad (3)$$

$$C_P = \frac{P}{\rho n^3 D^5} \quad (4)$$

For dynamic testing of propellers, the efficiency, η , of the propeller can be calculated by dividing the output power ($P_{out} = T \cdot V_\infty$) by the input power, P , to the motor. Simplifying in terms of the advance ratio, J , and non-dimensional coefficients yields Equation (5).

$$\eta = \frac{C_T J}{C_P} \quad (5)$$

Thus, with measured values for V_∞ , n , D , T , ρ , and P , the performance of a propeller can be fully characterized in non-dimensional parameters and readily compared to results from prior research.

Experimental Setup

Design Overview

The test stand was designed to provide a rigid test platform that would elevate the propeller above the roof of the car to mitigate aerodynamic boundary effects and allow the propeller to experience undisturbed air. The test stand construction consisted of two vertically-oriented steel legs, an aluminum square tube to which sensors were mounted, and steel braces set at an angle to provide stable footing. The length, L , of the stand's base was chosen to utilize all the usable space on each of the vehicle's rails. The stand was mounted to the vehicle using clamps designed to interface with the existing roof rack.

A basic schematic is presented in **Figure 1**. For further detail, see **Appendix C**.

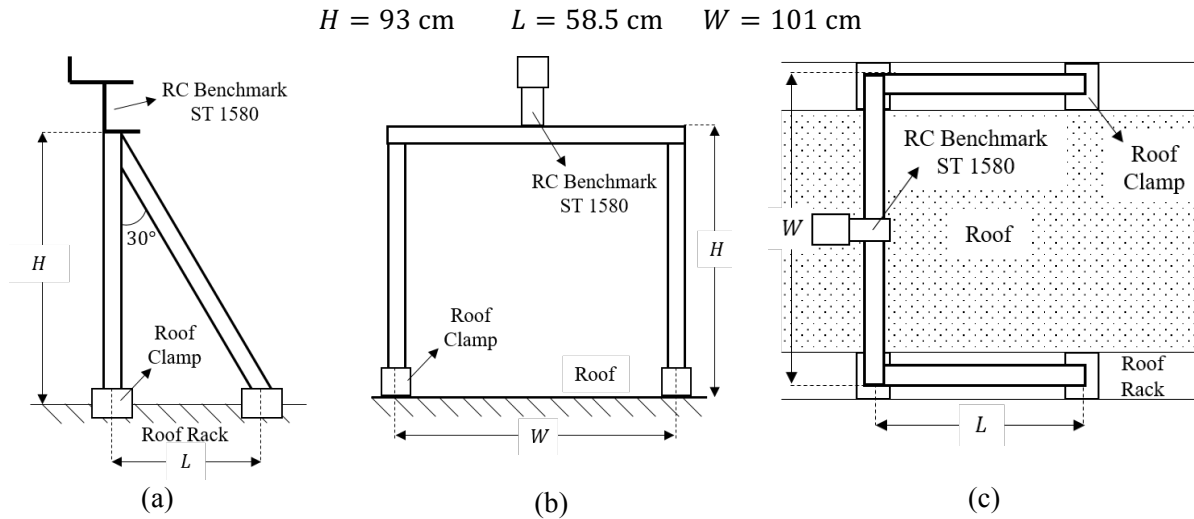


Figure 1. Side (a), front (b), and top view (c) of the resultant dynamic test stand that exceeds the minimum required height to assure appropriate air quality for propellers of up to 11 in and utilizes the entire roof rack length to maximize support and safety.

In order to construct an experimental setup that meets all of the design objectives and allows for quality data to be obtained, aerodynamics, air quality, and systems integration must be adequately addressed.

Aerodynamic Considerations

To ensure the entire propeller is immersed in a uniform freestream flow, the thickness of the boundary layer at the location of the propeller must be known. Approximating the surface of the vehicle (for this paper, a 2013 Subaru Outback) as a flat plate, the Reynolds number is given by Equation (6).

$$Re_x = \frac{\rho V_\infty x}{\nu} \quad (6)$$

where ρ is the density of the ambient air, V_∞ is the freestream velocity, x is the characteristic length in the flow-wise direction, and ν is the kinematic viscosity of the ambient air.

Assuming the flow along the surface to be initially turbulent, the boundary layer thickness, δ , is given by Equation (7). [4]

$$\delta = \frac{0.385}{Re_x^{0.2}} x \quad (7)$$

For the worst-case boundary layer thickness, conditions are assumed to be at sea level using the International Standard Atmosphere.

T	15°C
ρ	1.225 kg/m ³
U_∞	22.4 m/s
ν	$1.81 \cdot 10^{-5}$ m ² /s
x	2.79 m

Table 1. Assumed worst-case atmospheric test conditions.

Using the values from **Table 1** in conjunction with Equations (6) and (7), the expected boundary layer thickness is about 5.1 cm.

In their book *Low-Speed Wind Tunnel Testing*, Pope and Harper suggest that the ratio of propeller disk area to wind tunnel test section area, A_{prop}/A_{WT} , be no greater than 0.15 in order to obtain data of acceptable quality [2]. This rule has been reformulated in more recent testing such that the width of the test section must be at least five times the propeller diameter in order to avoid boundary interference effects on the performance of the propeller [1]. The proposed vehicle-mounted test stand is only constrained on one side (in contrast to the four walls of a traditional wind tunnel). Based on symmetry, the propeller's radius, r_{prop} , will be used in calculations in order to match the more accurately represented boundary conditions. Therefore, the minimum distance required between the center of the propeller hub and the roof of the vehicle is

$$H_{min} = 5 \cdot r_{prop} + \delta \quad (8)$$

The largest propeller tested had $D = 27.9$ cm (11 in), which gives $H_{min} = 74.92$ cm (29.5 in).

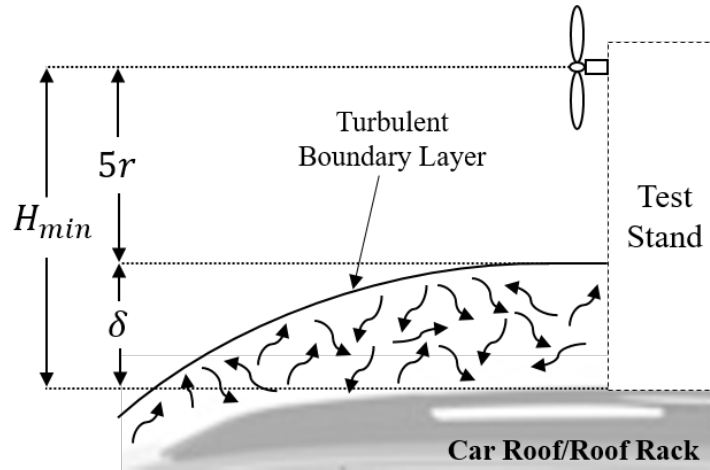


Figure 2. To provide the tested propeller with uncompromised air quality, the test stand must have a height of at least 75 cm to avoid interference by the turbulent boundary layer of the car and the boundary interference effects created by the roof of the car for testing a 28 cm (11 in) propeller.

In order to reduce manufacturing lead times, prefabricated 36 inch (91.44 cm) L-shaped steel rails were used for the vertical supports. As a result, the height of the test stand as built exceeded the minimum required height (H_{min}), compensating for any uncertainty in predictions and allowing larger diameter propellers to be tested.

Data System

The test system as designed required the use of two independent data systems. Consequently, special attention was given to the integration and testing of these systems with the test vehicle prior to data collection. A functional schematic and layout of the system is provided in **Figure 3**.

Data was gathered primarily through the use of RC Benchmark's S1580 dynamometer, which directly houses the motor and propeller. The dynamometer contains an onboard data acquisition board that provides performance data (thrust, torque, RPM, current, voltage, airspeed) in real-time to a laptop-based user interface via a USB connection. All data was logged and exported as a .csv file for post-processing.

Since the primary challenge in performing out-of-wind-tunnel testing is the unpredictability of the testing environment, an independent air quality monitoring system was also employed. This system consists of a Pitot-static tube for airspeed monitoring and a wind vane coupled with a rotary position sensor to monitor the presence of crosswinds. These sensors were integrated with a National Instruments data acquisition system and LabVIEW™ was used for data recording and real-time monitoring of test conditions. With the wind vane, it could be verified that the free stream was normal to the plane of the propeller, with an acceptable tolerance of 5% [5].

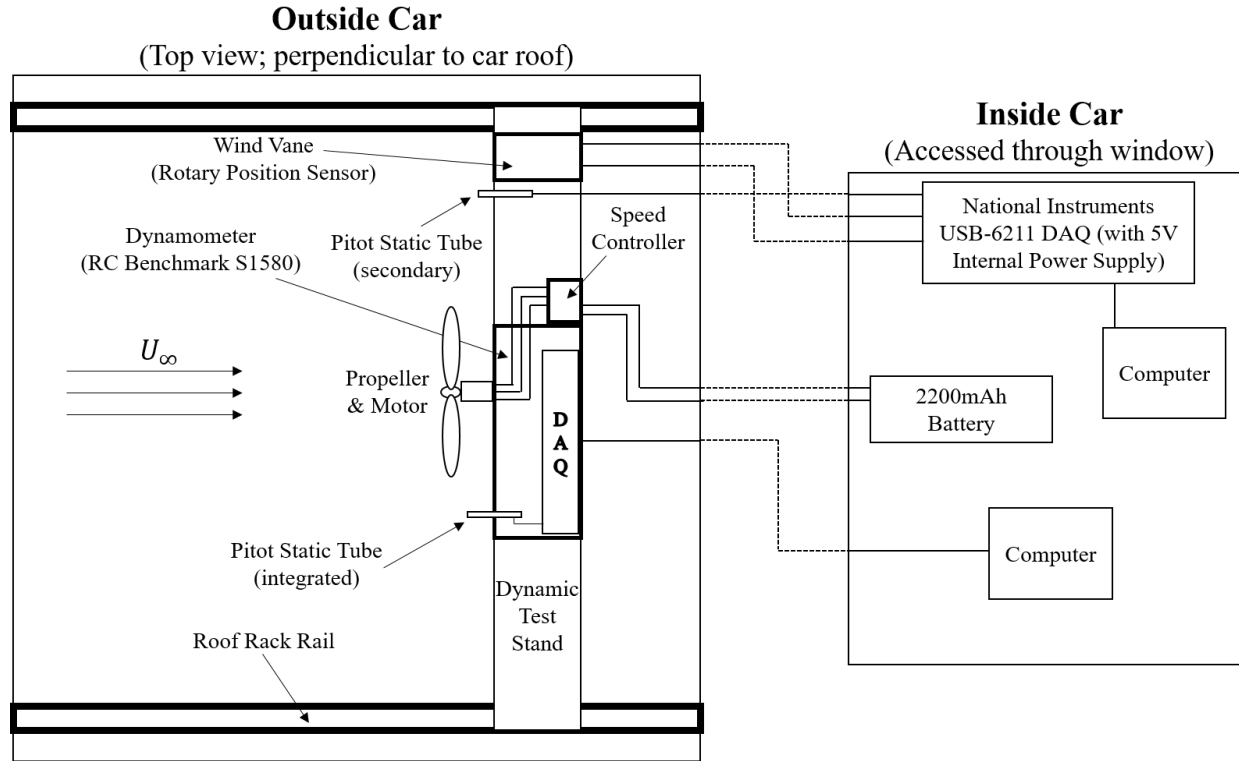


Figure 3. Layout and functional schematic of the experimental setup.

The specifications of the RC Benchmark S1580 dynamometer are provided in **Table 2**.

	Min.	Max.	Resolution	Unit
Thrust	-49	49	0.5%	N
Torque	-1.5	1.5	0.5%	N-m
Angular Speed	0	190,000*	< 1%	eRPM*
Airspeed	-105	105	1%	m/s
Voltage	0	35	0.5%	V
Current	0	40	1%	A

Table 2. RC Benchmark S1580 dynamometer specifications. *Mechanical RPM can be obtained by dividing eRPM by the number of motor poles

The specifications of the air quality monitoring system, which employed a differential pressure transducer (All Sensors 5 INCH D-4V) and a through-shaft rotary position sensor (Honeywell 640GS103B06NBAY), are provided in **Table 3**.

	Min.	Max.	Resolution	Unit
Free Stream Angle	0	360	5%	Degrees
Airspeed	-45.52	45.52	0.1%	m/s

Table 3. Air quality monitoring system instrument specifications

Aerodynamic Drag on Dynamic Test Stand

The exposure of the front profile of the test stand to the free stream, as seen in **Figure 1** (b), resulted in drag opposing the thrust produced by the propeller. As a result, thrust measurements produced by the appropriate load cells on the dynamometer were lower than the actual thrust produced by the propeller. To remedy this, a calibration run was conducted to characterize the drag on the dynamometer as a function of freestream velocity. A drag calibration curve was generated from this data, which was used to modify thrust measurements in post-processing to obtain the true thrust produced by the propeller.

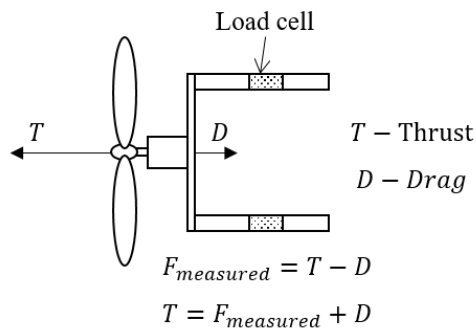


Figure 4. The measured thrust, $F_{measured}$, is the difference between the true thrust, T , and the drag of the test stand, D .

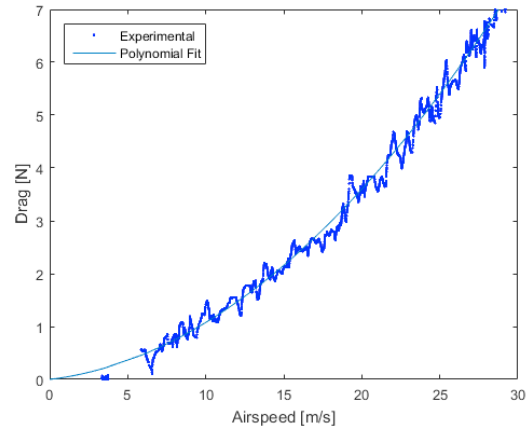


Figure 5. The drag, D as a function of V_{∞} allows the true thrust to be corrected for data post processing.

Test Procedure

Testing was performed at the El Mirage Dry Lake Off-Highway Vehicle Recreation Area in Palmdale, California. The test site was chosen for its long, smooth driving surface and the freedom to drive for several miles uninterrupted in any direction. These conditions were ideal for minimizing vibration of the test stand originating from the car as well as eliminating crosswinds.

Static Calibration

A static calibration of the system was performed before the start of each test session. The airspeed sensors on both the dynamometer and air quality monitor system were tared separately by covering the static and dynamic ports on each respective pitot tube. Then, the wind vane was tared by physically rotating it to its predetermined zero-degree position. After performing the manufacturer procedure for calibrating the load cells of the dynamometer, both the thrust and torque readings were tared using the software interface provided by the manufacturer.

Dynamic Calibration

Following the static calibration of the system, a dynamic calibration was performed to directly measure the drag on the dynamometer. This was done by mounting the complete system to the vehicle without a propeller and recording the force on the dynamometer continuously while traveling from 5 MPH to 80 MPH, holding velocity steady for 5 seconds every 5 MPH. The top speed of 80 MPH was chosen due to the maximum required speed to achieve windmill state (zero-thrust condition) for the largest propeller to be tested, per Equation (2).

Measurement Procedure

Data was collected for a range of propeller diameters and pitches in order to quantify the repeatability and quality of data collected over a range of geometries. Additionally, tests were conducted to characterize the propeller size limitations inherent to the test stand. The test matrix is summarized in **Table 4**.

APC 6x5.5 @ 7000 RPM		
APC 9x7.5 @ 5000 RPM	APX 9x9 @ 5000 RPM	
APC 11x5.5 @ 5000 RPM	APC 11x7 @ 6000 RPM	APC 11x8 @ 5000 RPM

Table 4. Summary of propellers tested.

For each test, a propeller was mounted on the stand and the throttle setting was set through an electronic speed controller to achieve the desired RPM. Holding RPM constant, data was recorded continuously while the vehicle was slowly accelerated in the same manner as the dynamic calibration until the windmill state was reached. This process was repeated for each propeller.

After each run, the drag force on the dynamometer was added back in to the measured thrust in order to attain the true thrust generated by the propeller. Furthermore, all non-dimensional performance coefficients (per Equations (3), (4), and (5)) were calculated to produce plots of C_T vs. J , C_P vs. J , and η vs. J . This allowed for evaluation of experimental data by direct comparison to theoretical data provided by Selig.

Results and Discussion

Test Repeatability

The data for repeated runs with the same propeller were compared to first evaluate the repeatability of each test. A representative plot is shown of the APC 9x7.5E propeller that compares the data collected across two separate runs with an identical experimental setup.

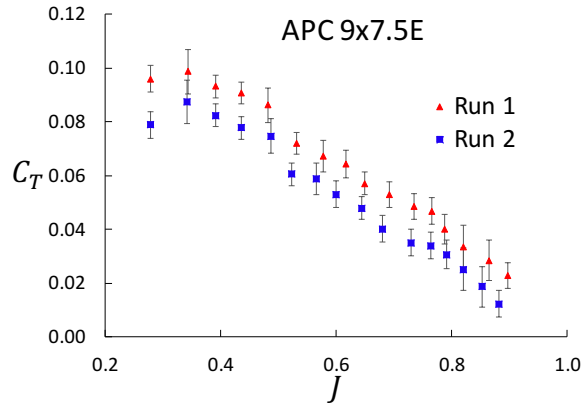


Figure 6. A comparison of two runs indicates that measured values of C_T are equivalent for nearly all values of J .

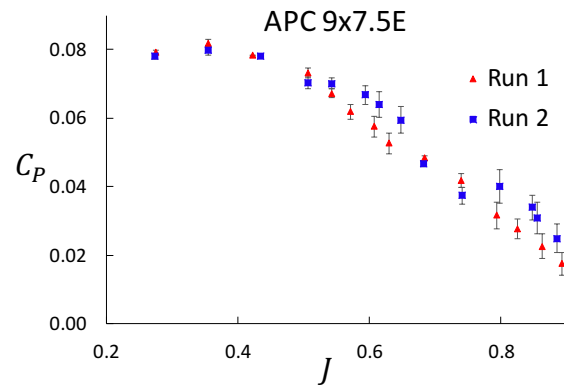


Figure 7. Comparing two runs indicates the measured values of C_P are equivalent for all values of J .

Figure 6 and **Figure 7** show matching between two separate tests of an APC 9x7.5E propeller, indicating that the dynamic test stand collects data with reliable consistency. This matching between test sets with the same conditions persists throughout the experimental data from all propellers tested. Because of the reliability of the data system and test stand measurements, it is reasonable to continue to a comparison of wind tunnel and out-of-wind-tunnel data.

Comparison with Literature Wind Tunnel Data

Following the validation of the test stand's measurement repeatability and consistency, experimental data was compared with “theoretical” data taken from wind tunnel tests performed by Deters and Selig [6].

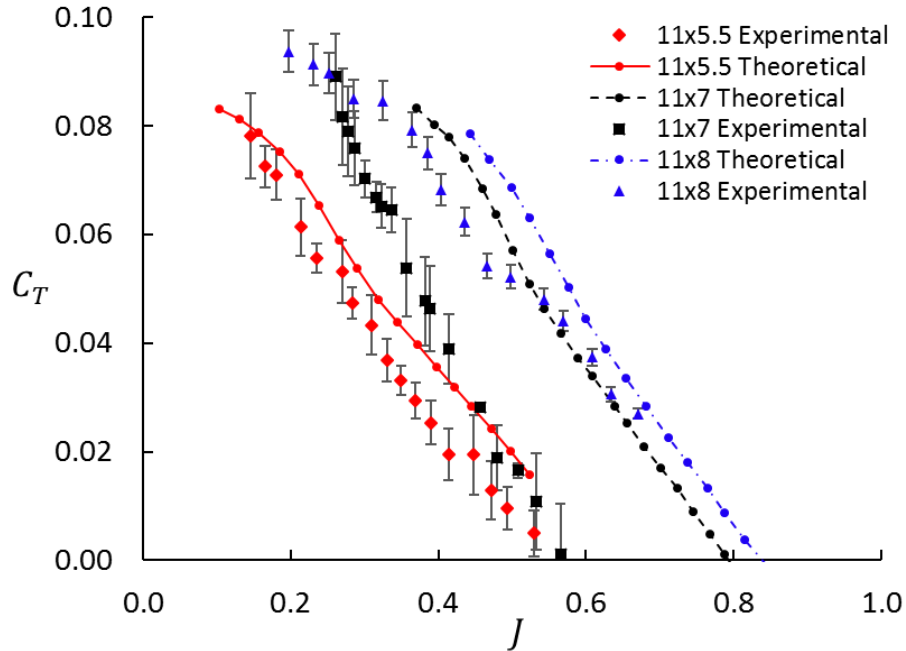


Figure 8. The magnitude of experimental C_T is consistently less than the “theoretical” literature data equivalent for any propeller and for all values of J .

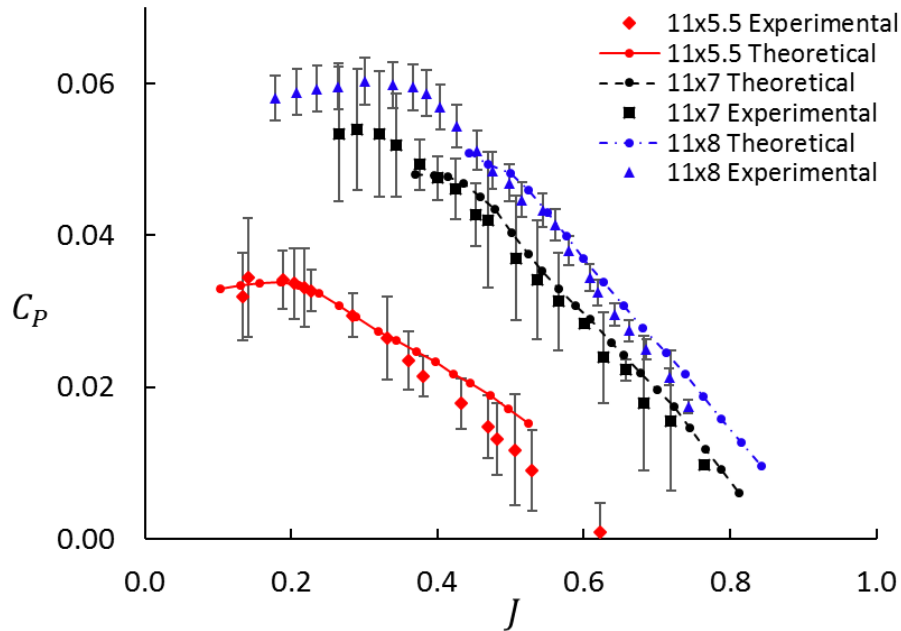


Figure 9. The magnitude of experimental C_P is equivalent to the “theoretical” data for all values of J .

As shown in **Figure 8**, the experimental data collected does not match theoretical values for C_T . The experimental data does, however, match the theoretical values for C_P , which is shown in **Figure 9**. This result is consistent with the discrepancy seen in previous iterations of Selig's tests, which were conducted by Brandt and are shown in **Figure 10** and **Figure 11** [7]. The inconsistency results from blockage effects created by the motor housing and front plate of the dynamometer. These blockage effects reduce the change in momentum across the propeller disc and result in lower measured thrust values. As the quantity C_P is derived independent of the measured thrust, the values for C_P remain unaffected by the blockage effects.

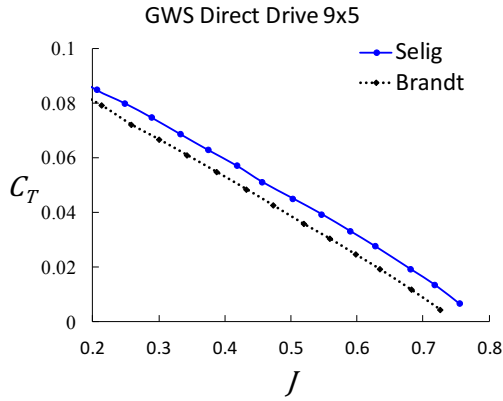


Figure 10. Brandt's results for C_T are consistently lower at any J than Selig's results.

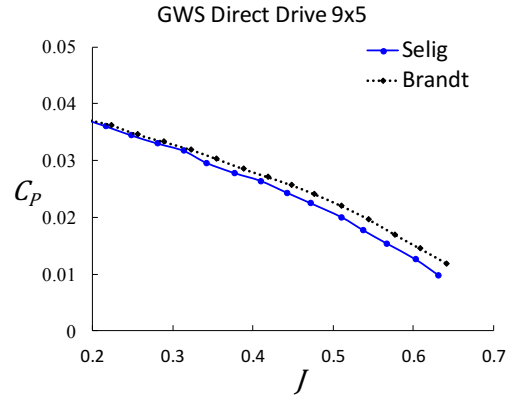


Figure 11. Brandt's results for C_P are slightly lower than Selig's results for $J > 0.35$.

The blockage effects propagate through to the calculated propulsive efficiency, η , as shown in **Figure 12** and **Figure 13**.

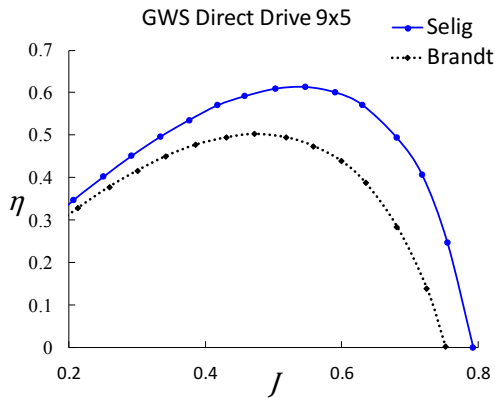


Figure 12. Due to consistently lower C_T , Brandt's results for η are also consistently lower than Selig's.

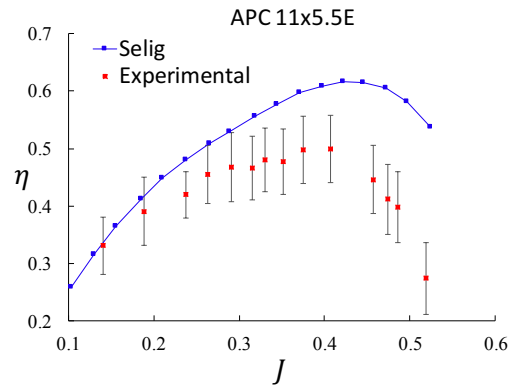


Figure 13. Experimental results for η from the test stand have a similar under-performing trend to Brandt's with respect to Selig's results.

It is important to note that Brandt's data is from a GWS Direct Drive 9x5 propeller, while the experimental data is from an APC 11x5.5E propeller. Furthermore, the magnitudes of thrust loss are not expected to match because of differences in experimental setups. However, the similarity in underperformance with respect to Selig's data is a result of the physical phenomenon of momentum loss behind the propeller in each experimental setup. As the test stand produces curves consistent with physical phenomena seen in prior research, it can be concluded that the test stand could be redesigned to appropriately mitigate blockage effects.

Limitations of the Experimental Setup

All prior experimental results were collected using APC 11 inch propellers. As shown in **Figure 14** and **Figure 15**, the 9 inch propellers tested have an offset in C_T and are equivalent to Selig data for C_P . This is consistent with the trends shown by the 11 inch propellers.

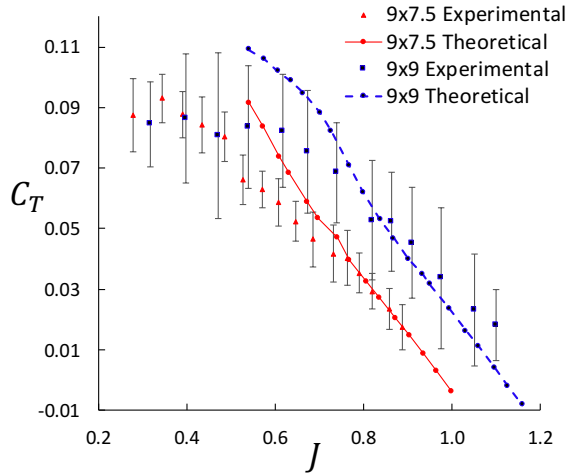


Figure 14. The 9 in APC propellers demonstrate trends consistent with the 11 in propellers.

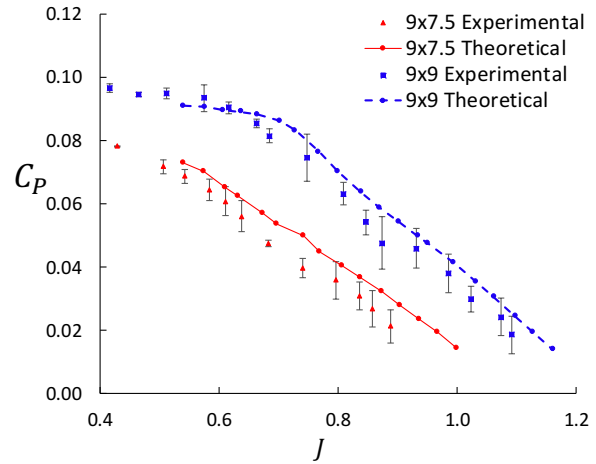


Figure 15. The plots of C_P for 9 in propellers are equivalent to the experimental Selig data.

Despite the success in measuring dynamic behavior of 9 and 11 *in.* propellers, the data collected for smaller propellers was not consistent with expectations from literature. As shown in **Figure 16**, C_T for a 6 in propeller did not decrease toward zero with increasing advance ratio. Additionally, test runs conducted with smaller propellers exhibited much less repeatability, which resulted in large uncertainty when compared to uncertainty for larger propeller runs.

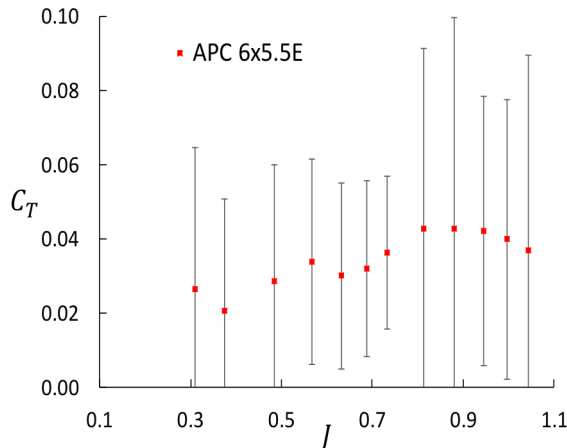


Figure 16. The thrust measurements for 6 inch propellers did not match expected trends from literature data.

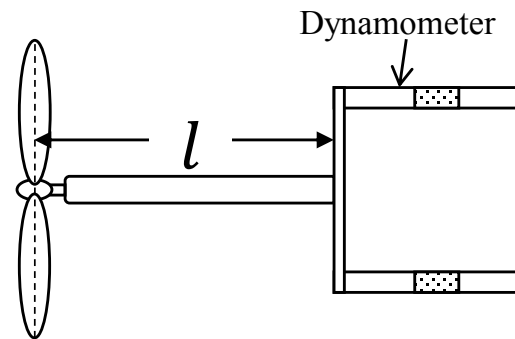


Figure 17. The distance, l , between the propeller and the dynamometer mount could be increased to reduce the momentum loss behind the propeller disc in future testing.

As a result of these observations, it was concluded that the magnitude of the noise from the measurement system was much closer to the magnitude of the signal associated with the thrust produced by the propeller, resulting in the trends in **Figure 16**. Although the discrepancy in C_T could similarly be attributed to momentum loss as a result of blockage effects behind the propeller disc, it is difficult to identify the primary cause from the data collected for this particular propeller because of a lower signal-to-noise ratio caused by lower thrust produced. Further testing could work to isolate and identify the root cause of the low-quality data shown in **Figure 16**. Because of the magnitude of the noise associated with data from small propellers, testing was primarily focused on higher diameter propellers that provided a much better signal-to-noise ratio. Based on the data collected during this experiment, it is recommended that this test stand be used for characterizing propellers of $D > 9$ in.

Conclusions

Experimental data collected using this dynamic thrust stand indicates that it can reliably characterize the dynamic performance of a propeller larger than 9 inches in diameter. Further improvements can be implemented that will reduce the effect of momentum losses and collect data consistent with equivalent wind tunnel tests. With a single day of testing, a UAV designer could catalog the behavior of a multitude of propellers under consideration for use in an aircraft's propulsion system. By comparing the propulsive efficiency of multiple propellers at different airspeeds and rotation rates, the best propeller can be selected for the aircraft's operating conditions. Testing that was previously only available to large universities can now be done without a wind tunnel and without the need for an expensive and time-consuming test flight.

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Appendix A – Uncertainty Analysis

$$J = \frac{V_{\infty}}{nD}$$

$$\frac{\partial J}{\partial V_{\infty}} = \frac{1}{nD}$$

$$\frac{\partial J}{\partial n} = \frac{V_{\infty}}{n^2 D}$$

$$\frac{\partial J}{\partial D} = \frac{V_{\infty}}{nD^2}$$

$$\Delta J = \sqrt{\left(\frac{\partial J}{\partial V_{\infty}} \Delta V_{\infty}\right)^2 + \left(\frac{\partial J}{\partial n} \Delta n\right)^2 + \left(\frac{\partial J}{\partial D} \Delta D\right)^2}$$

$$\Delta J = \sqrt{\left(\frac{\Delta V_{\infty}}{nD}\right)^2 + \left(\frac{V_{\infty} \Delta n}{n^2 D}\right)^2 + \left(\frac{V_{\infty} \Delta D}{nD^2}\right)^2}$$

$$C_T = \frac{T}{\rho n^2 D^4}$$

$\frac{\partial C_T}{\partial T} = \frac{1}{\rho n^2 D^4}$	$\frac{\partial C_T}{\partial \rho} = \frac{T}{\rho^2 n^2 D^4}$	$\frac{\partial C_T}{\partial n} = \frac{2T}{\rho n^3 D^4}$	$\frac{\partial C_T}{\partial D} = \frac{4T}{\rho n^2 D^5}$
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$$\Delta C_T = \sqrt{\left(\frac{\partial C_T}{\partial T} \Delta T\right)^2 + \left(\frac{\partial C_T}{\partial \rho} \Delta \rho\right)^2 + \left(\frac{\partial C_T}{\partial n} \Delta n\right)^2 + \left(\frac{\partial C_T}{\partial D} \Delta D\right)^2}$$

$$\Delta C_T = \sqrt{\left(\frac{\Delta T}{\rho n^2 D^4}\right)^2 + \left(\frac{T \Delta \rho}{\rho^2 n^2 D^4}\right)^2 + \left(\frac{2T \Delta n}{\rho n^3 D^4}\right)^2 + \left(\frac{4T \Delta D}{\rho n^2 D^5}\right)^2}$$

$$C_P = \frac{P}{\rho n^3 D^5}$$

$\frac{\partial C_P}{\partial P} = \frac{1}{\rho n^3 D^5}$	$\frac{\partial C_P}{\partial \rho} = \frac{P}{\rho^2 n^3 D^5}$	$\frac{\partial C_P}{\partial n} = \frac{3P}{\rho n^4 D^5}$	$\frac{\partial C_P}{\partial D} = \frac{5P}{\rho n^3 D^6}$
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$$\Delta C_P = \sqrt{\left(\frac{\partial C_P}{\partial P} \Delta P\right)^2 + \left(\frac{\partial C_P}{\partial \rho} \Delta \rho\right)^2 + \left(\frac{\partial C_P}{\partial n} \Delta n\right)^2 + \left(\frac{\partial C_P}{\partial D} \Delta D\right)^2}$$

$$\Delta C_P = \sqrt{\left(\frac{\Delta P}{\rho n^3 D^5}\right)^2 + \left(\frac{P \Delta \rho}{\rho^2 n^3 D^5}\right)^2 + \left(\frac{3P \Delta n}{\rho n^4 D^5}\right)^2 + \left(\frac{5P \Delta D}{\rho n^3 D^6}\right)^2}$$

$$\eta = \frac{C_T J}{C_P}$$

$$\frac{\partial \eta}{\partial C_T} = \frac{J}{C_P}$$

$$\frac{\partial \eta}{\partial J} = \frac{C_T}{C_P}$$

$$\frac{\partial \eta}{\partial C_P} = \frac{C_T J}{C_P^2}$$

$$\Delta \eta = \sqrt{\left(\frac{\partial \eta}{\partial C_T} \Delta C_T\right)^2 + \left(\frac{\partial \eta}{\partial J} \Delta J\right)^2 + \left(\frac{\partial \eta}{\partial C_P} \Delta C_P\right)^2}$$

$$\Delta \eta = \sqrt{\left(\frac{J \Delta C_T}{C_P}\right)^2 + \left(\frac{C_T \Delta J}{C_P}\right)^2 + \left(\frac{C_T J \Delta C_P}{C_P^2}\right)^2}$$

Appendix B – Risk Assessment & Alleviations

Test Location

El Mirage Dry Lake Off-Highway Vehicle Recreation Area, San Bernardino County
Managed by the Bureau of Land Management (BLM)

General Site Rules

- There is no speed limit on the open lakebed
 - The speed limit is 15 mph within 50 feet of campsites and staging areas
 - Camping is restricted to within 100 feet of the edge of the lakebed
- The dry lakebed is closed when wet or muddy
- Headlights and taillights are required when driving between one-half hour after sunset to one-half hour before sunrise
- Motorized vehicles must yield to non-motorized vehicles

Risks

- Encountering other vehicles
- Test stand detachment from vehicle or propeller detachment from test stand

Risk Alleviations

- Testing will be conducted during off-peak times (early weekday mornings 6am to 11 am) in order to minimize the likelihood of encountering other vehicles.
- One member of the test crew will be solely focused on monitoring the vehicle's surroundings
- Any member of the test crew will have full authority to halt testing at any time should they deem any portion of the test conditions or environment unsafe.
- Each bolt on the test stand will be tightened before starting each phase of testing. Motor and propeller mounts will be checked each time they are changed.

Test Procedure

1. Prior to each test condition, tighten each bolt on the test stand. Shake the stand at several points to ensure no component is loose.
2. Check that all cables are properly secured.
3. Tighten bolts on motor mount and propeller
4. Brief condition to be tested, making note of and verifying:
 - a. Motor Type
 - b. Propeller Size
 - c. Desired RPM
 - d. Starting Velocity
 - e. Velocity Increase Step Size
 - f. Expected Maximum Velocity
5. Load data acquisition software and tare sensors
6. Record initial wind data prior to start of test condition
7. Conduct test condition as briefed
8. Decelerate to a full stop and place the car in park
9. Record final wind data after completion of test condition

Crew Roles and Responsibilities

The test crew shall consist of no less than three members. All test participants will be briefed on the testing to be conducted on the test day in question prior to the start of testing. Additionally, all test participants will be familiar with the crew roles and responsibilities outlined in this document.

The roles of each crew member shall be as follows:

1. Driver

The driver will be responsible for operation of the vehicle. Their sole focus will be monitoring the environment directly ahead of the vehicle for any potential hazards. Secondary duties will be maintaining the vehicle's course (steering) and speed (cruise control).

2. Safety Focal

The duty of the safety focal will be to monitor the testing environment to identify potential safety hazards and promptly communicate relevant status updates to the driver. The safety focal shall have full authority to halt testing at any time should they deem the conditions unsafe.

3. Data Analyst

The data analyst will be responsible for monitoring data from both data collection systems. Their primary focus will be on monitoring the quality of data from the dynamometer and communicating to the driver when to proceed to the next test condition. Their secondary focus will be on monitoring the velocity of the free stream through the secondary data acquisition unit, ensuring the airspeed is as required per condition and that the wind direction does not deviate more than five (5) degrees from the vehicle's direction of travel. This duty shall be the sole focus of the Condition Quality Monitor if a fourth crew member is present.

4. Condition Quality Monitor (Optional)

If a fourth crew member is present, their sole focus will be on monitoring the velocity of the free stream through the secondary data acquisition unit, ensuring the airspeed is as required per condition and that the wind direction does not deviate more than five (5) degrees from the vehicle's direction of travel. The option for a fourth member is included in order to reduce the workload of the rest of the crew.

Emergency Contact Information

BLM Emergency Assistance
(888) 233-6518

BLM Barstow Field Office/Field Permit Information
(760) 252-6000

El Mirage OHV Area Recorded Information
(760) 252-6011

St. Mary Regional Medical Center
18300 Highway 18, Apple Valley, CA
(760) 242-2311

Victor Valley Community Hospital
15248 11th Street, Victorville, CA
(760) 245-8691

References

Bureau of Land Management, El Mirage Dry Lake OHV Recreation Area, Accessed September 6, 2016, [<http://www.blm.gov/ca/st/en/fo/barstow/mirage.html>]

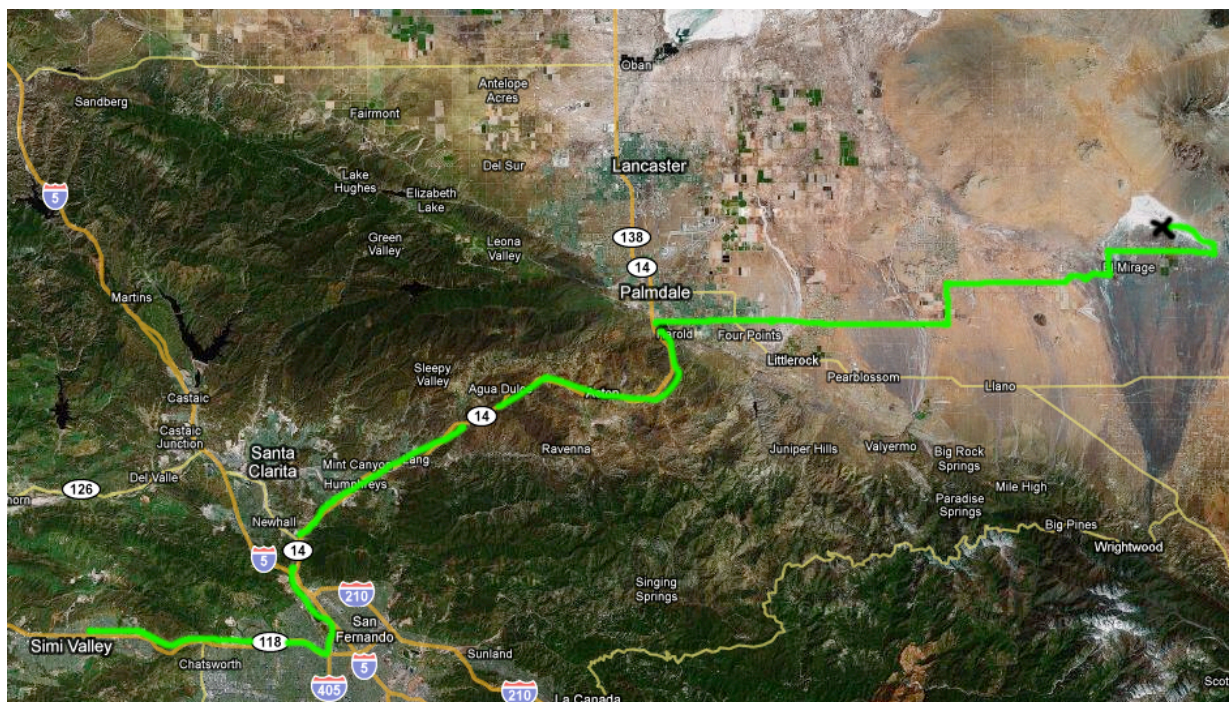


Figure 18. Location of El Mirage Dry Lake OHV Recreation area with driving directions from Los Angeles.

Appendix C – Technical Drawings

Drawings are provided here to allow the reader to recreate the test stand for their own use. For components that were not custom-made, see the part numbers listed in the itemized project budget in Appendix E.

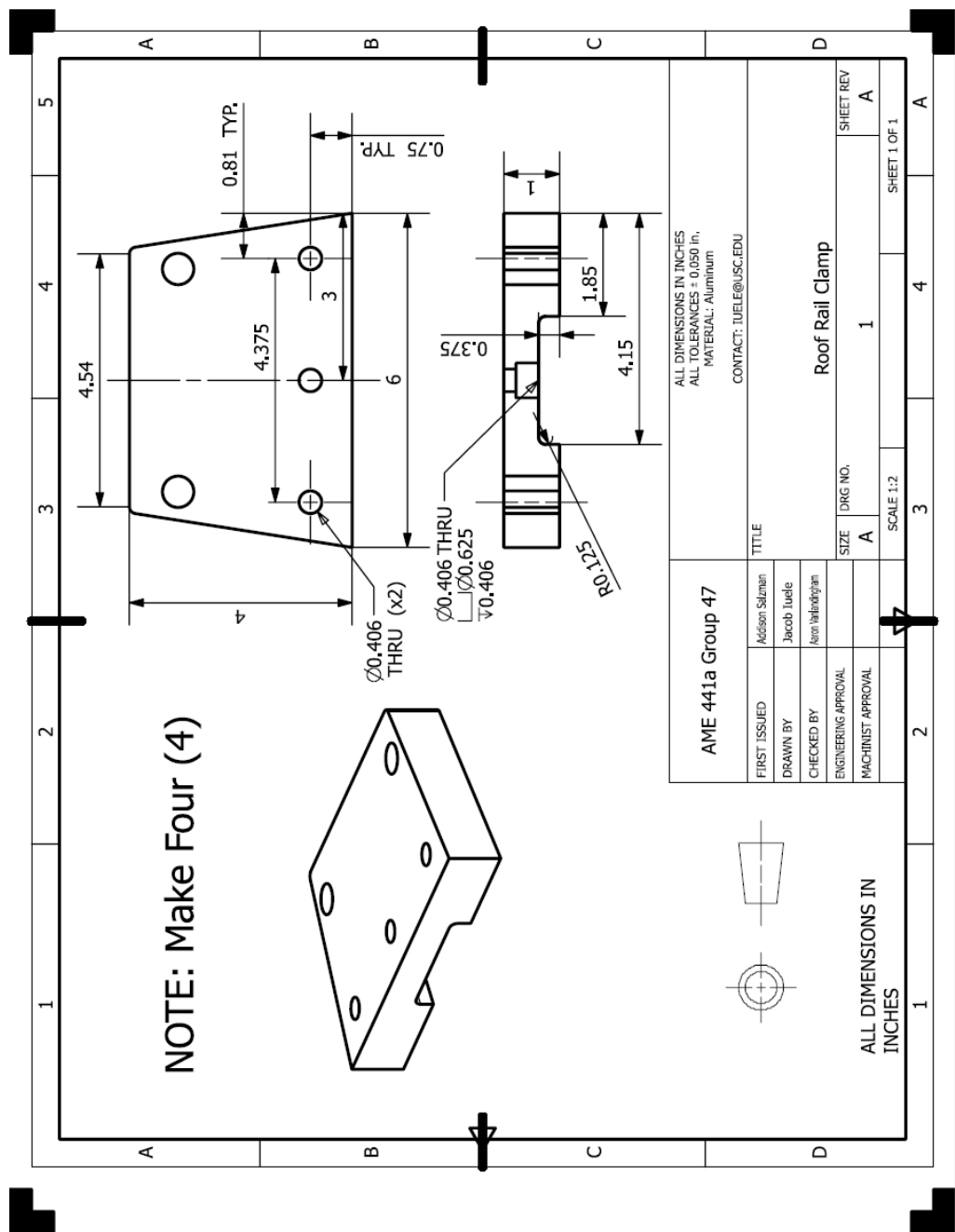


Figure 19. The top half of the clamp used to secure the stand to the test vehicle. The clamp was attached to the stand via a 90-degree angle bracket and a socket-head cap screw placed in the counter-bored hole. Clamping force was provided by a 4" U-bolt and the bottom half of the clamp was made of wood and intended only to distribute the clamping force more evenly.

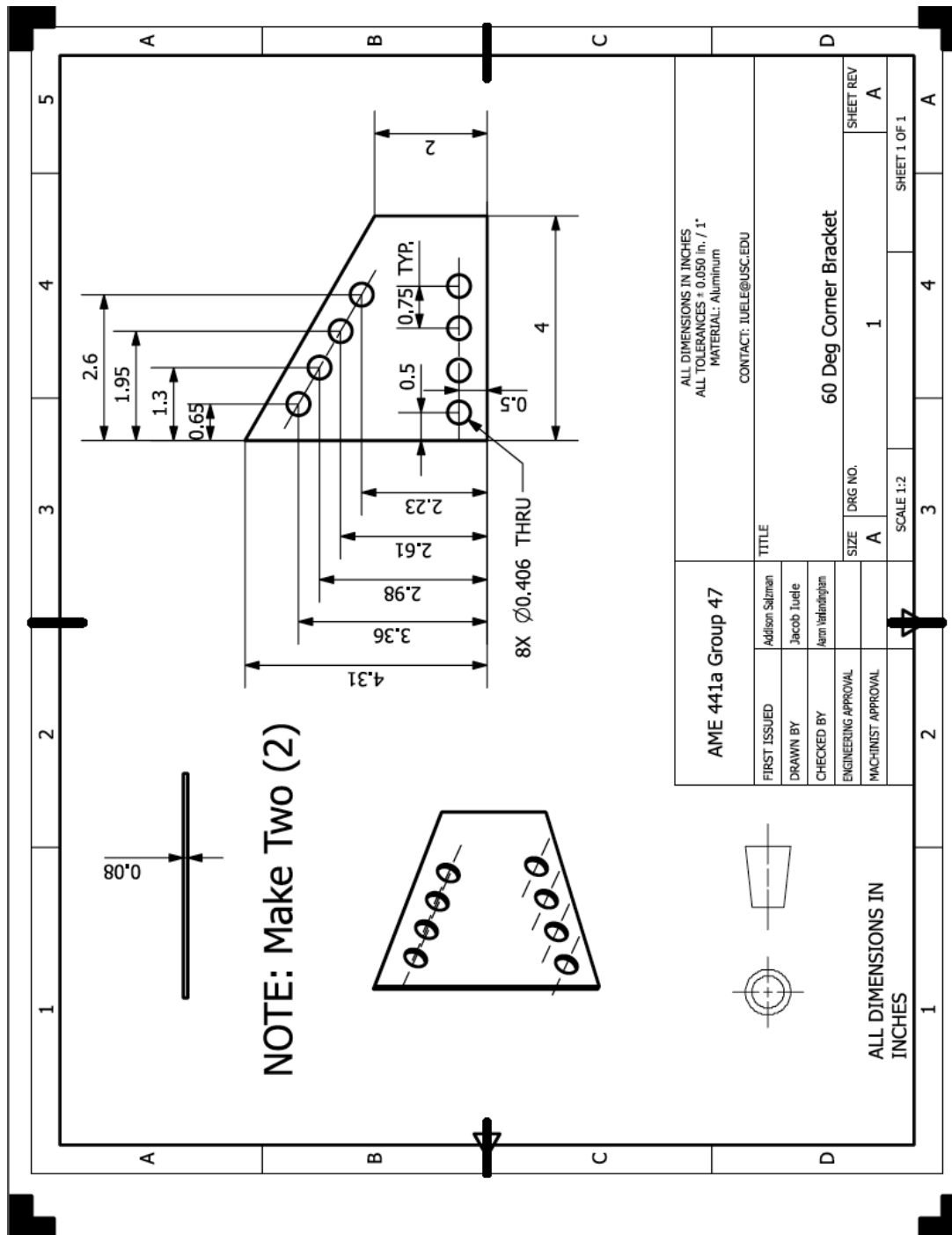


Figure 20. 60-degree angle bracket designed to interface the vertical supports with the angled braces. Holes are provided to match the size and spacing of the prefabricated parts (McMaster part no. 8968K24). Components were assembled using 1-inch 3/8-16 cap screws.

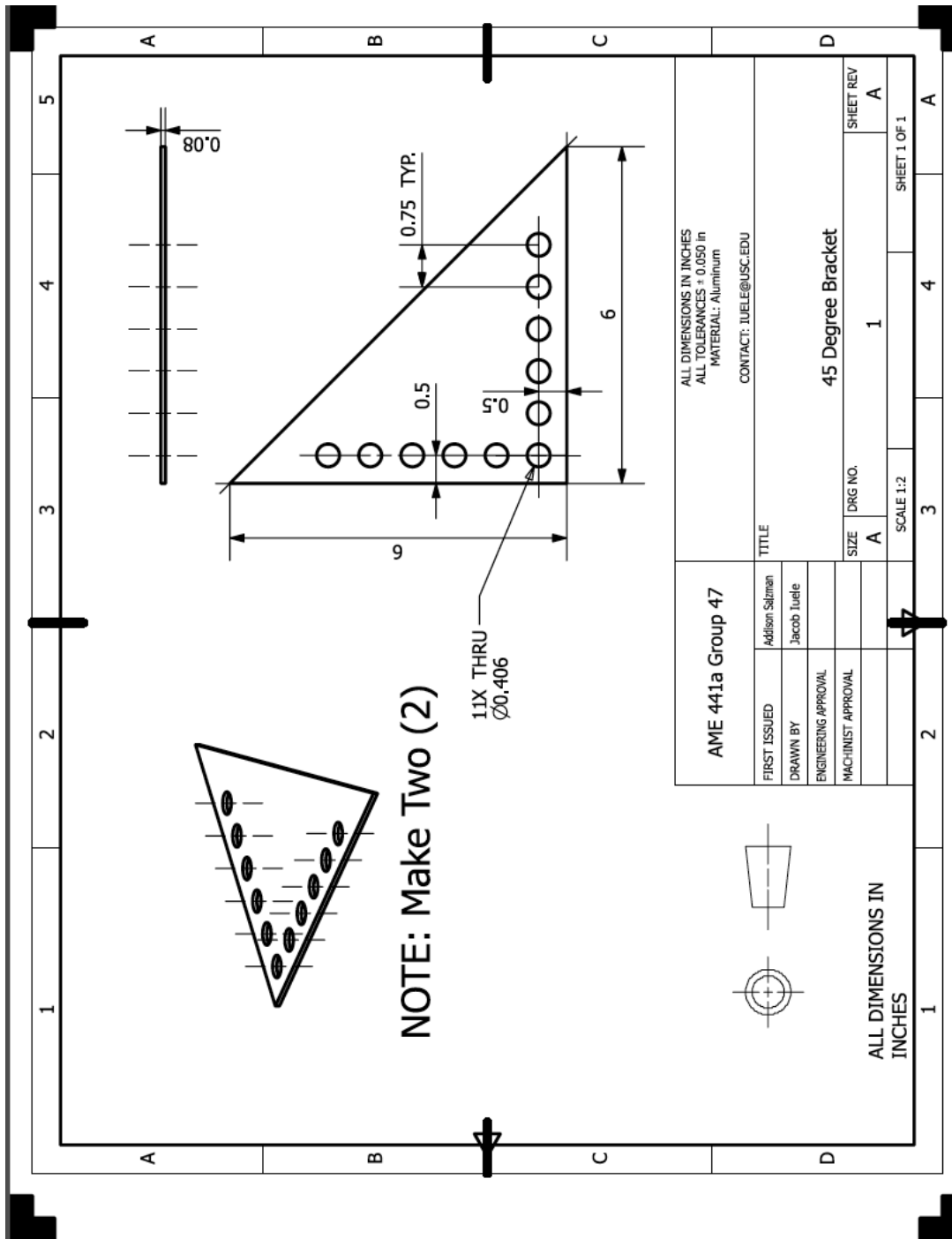


Figure 21. 45-degree angle bracket designed to be the interface between the vertical supports (McMaster part no. 8968K24) and horizontal square tube (McMaster part no. 8809T11) to which sensors were mounted.

Appendix D – Project Timeline

AME-441 Group 47 - Timetable
Out-of-Wind-Tunnel Dynamic Testing of Propellers

Aaron VanLandingham	AV
Addison Salzman	AS
Jacob Iuele	JI



Appendix E – Project Budget and Cost

The allocated budget for the project was \$75/student, for a total of \$225. Through the use of university resources, total expenses for the project were limited to \$204.37.

AME-441 Group 47 - Cost Estimate and Expenses Out-of-Wind-Tunnel Dynamic Testing of Propellers Aaron VanLandingham, Addison Salzman, Jacob Iuele					
	Vendor/Manufacturer	Model/Part No.	Qty.	Unit Price	Item Price
EXPENSES FOR ITEMS AVAILABLE IN LAB					
Equipment					
USB Multifunction DAQ	National Instruments	USB-6211	1	\$792.00	\$792.00
Vehicle for Mounting Testbed	Subaru	2013 Outback	1	\$15,000.00	\$15,000.00
Pressure Transducer	Digi-Key	MPXV7002DPT1TR-ND	1	\$8.20	\$8.20
Pitot-Static Tube	Eagle Tree Systems	B0023EJQFG	1	\$13.22	\$13.22
Lithium-Polymer Battery	Turnigy	2200 mAh 3S 20C	2	\$18.14	\$36.28
Materials					
Galvanized Low-Carbon Steel (90 Deg, Perforated, 1-1/2"x1-1/2"x3")	McMaster-Carr	8968K24	2	\$4.94	\$9.88
Galvanized Low-Carbon Steel (90 Deg, Perforated, 1-1/2"x1-1/2"x6")	McMaster-Carr	8968K24	3	\$8.66	\$25.98
Steel Cap Screws (1/4"-20, Full Thread, Length 2", QTY 50)	McMaster-Carr	92865A549	1	\$12.04	\$12.04
Hex Nut (1/4"-20, QTY 100)	McMaster-Carr	95462A029	1	\$4.40	\$4.40
Manufacturing					
Milling (\$70 per hour)	AME-441 Machine Shop		8	\$70.00	\$560.00
EXPENSES FOR ITEMS ON LOAN FROM USC AERODESIGN TEAM					
Dynamometer	RC Benchmark	S1580	1	\$495.00	\$495.00
Phoenix Ice 75 Speed Controller	Castle Creations	010-0067-00	1	\$119.95	\$119.95
Brushless Motor	Hacker	A30-14 L V3	1	\$81.49	\$81.49
Thin Electric Propellers	APC Propellers		10	\$2.39	\$23.90
				Sub-total Cost:	\$16,462.00
EXPENSES FOR ITEMS REQUIRED FOR PURCHASE					
Equipment					
Airspeed Sensor Kit	RC Benchmark		1	\$58.05	\$58.05
Honeywell Rotary Position Sensor	Digi-Key	480-5912-ND	1	\$24.45	\$24.45
25' USB Extender/Repeater	Amazon	B00FFKIUIS	1	\$9.79	\$9.79
Materials					
Square U-Bolt, 4" Inside	McMaster-Carr	3060T48	8	\$3.13	\$25.04
Steel Cap Screws (3/8"-16, Full Thread, Length 1", QTY 50)	McMaster-Carr	92865A624	1	\$10.37	\$10.37
Grade 5 Hex Nut (3/8"-16, QTY 100)	McMaster-Carr	95462A031	1	\$7.84	\$7.84
Strut Channel Connector (60 Degree, Outside Angle)	McMaster-Carr	33125T46	4	\$2.19	\$8.76
Aluminum Sheet, 12"x12"x0.08"	McMaster-Carr	89015K16	1	\$19.29	\$19.29
Aluminum Square Tube, 1-1/2"x4'	McMaster-Carr	8809T11	1	\$40.78	\$40.78
				Sub-total Cost:	\$204.37
TOTAL PROJECT EXPENSES:					\$16,666.37
BUDGET					
AME-441 Project Budget (per student)			3	\$75.00	\$225.00
				TOTAL BUDGET:	\$225.00
BUDGET REMAINING					\$20.63